2. Hot rolled cold twisted deformed (CTD) bars like Tor steel bars. High strength is achieved by cold twisting.

3. Thermo mechanically treated (TMT) bars. High strength is achieved by controlled cooling.

2.7.2 Rolled Sections- Structural Steel

The two main types of structural steel members are:

(i) The conventional hot-rolled steel sections.

(ii) Cold- formed steel sections.

(i) Hot- rolled steel sections

Steel used for fabrication of trusses, column, beams, etc., is made by rolling hot steel into various shapes in specially designed rolling mills. Hot rolled steel sections are illustrated in figures. The various sections available are as follows:

1. **Angle sections:** Various sizes of equal and unequal angle sections are available. They are mainly used for truss work and filler joist floors.

2. **Channel sections:** Bureau of Indian Standards classifies channels as junior channels (ISJC), light channels (ISLC) and medium channels (ISMC).They are used widely for steel framed structures.

3. **I sections:** These sections are called rolled steel joists or beams. They are classfied as Indian standard junior beam (ISJB), Indian standard light beam (ISLB), Indian standard medium beam (ISMB), Indian standard wide flanged beams (ISWB) and Indian standard heavy beam (ISHB). They are used in multistoried buildings, bridge and other places where bending stresses are very high.

4. **T sections:** These sections are used to make built up sections and roof girders.

5. **Other rolled sections:** Plain sheets, corrugated sheets, plates, expanded metal, sheet piles, rail sections, flats of varying width and thickness are also rolled in rolling mills. They are used for fabrication.



(ii) Cold-formed steel sections

Cold formed light gauge steel sections are structural members, cold formed to the desired structural shapes from carbon or alloy steel (strips or flats)by press- brake operations. The thickness of the member ranges from 0.38 to 6.35 mm. They have much higher strength than hot-rolled sections.

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